ANALYSIS TOOLS

01 CORE ANALYSIS

THE SAME 12 EXAMPLE RESTAURANTS

Region	Service time (seconds)	Matching CSAT (0-5)
Central	60, 68, 76, 84, 92, 100	3.5, 3.3, 3.4, 3.0, 3.1, 2.9
Suburban	110, 118, 126, 134, 142, 150	4.5, 4.3, 4.4, 4.0, 4.1, 3.9

1 feature-target-associations

Tests each feature against CSAT separately to find relationships worth exploring.

Staff per restaurant in each region: 6, 5, 4, 3, 2, 1



More staff and longer service times each correlate with higher CSAT here. Neither result adjusts for region.

2 target-by-numeric-bins

Groups service times into ranges, then compares average CSAT across those ranges.

3 equal-count time bins; 4 restaurants per bin



Slower bins have higher CSAT because they mix regions. This does not mean making service slower improves satisfaction.

3 target-by-group

Compares average CSAT across categories, such as restaurant regions.

12-restaurant calculation; mean CSAT

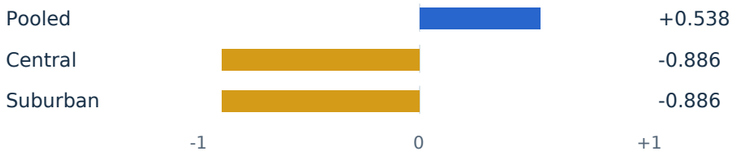


Suburban averages 1.0 CSAT point above Central. This is a regional gap without adjustment for service time.

4 correlation-by-group

Checks the service time-CSAT relationship separately inside each region.

12-restaurant calculation; Spearman r

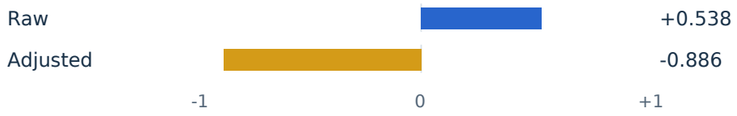


The direction reverses: within each region, longer service times go with lower CSAT. Pooling the regions hides this.

5 adjusted-correlation

Measures a relationship after removing associations with specified controls.

Control: region; calculation on ranks

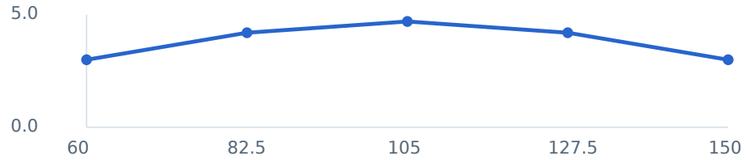


After region adjustment, longer service times correlate with lower CSAT. This matches tool 4's combined within-group r.

6 nonlinearity-check

Checks whether a curve explains a relationship that a single correlation can miss.

Separate illustration; service seconds vs CSAT

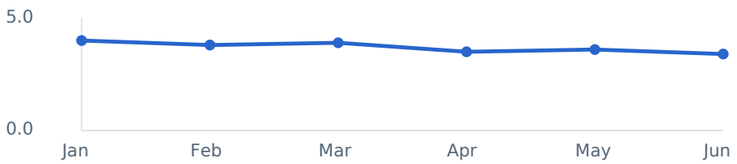


This separate example peaks at 105 seconds. Both shorter and longer times have lower CSAT; Spearman r = 0 misses the curve.

7 time-trend

Groups observations by month and shows how average CSAT moves over time.

Listed restaurant order: Jan-Jun in each region



CSAT generally falls (r = -0.886 with time). Different restaurants appear each month, so this is not same-store deterioration.

8 predictive-feature-importance

Fits features together; shuffles each one to see how much CSAT prediction worsens.

Illustrative scores; not fitted to these 12 restaurants



Illustration: shuffling service time lowers holdout R-squared from 0.80 to 0.50. The 0.30 drop shows predictive value, not causation.

9 multiple-testing-adjustment

Adjusts p-values across unique tests so a large search is less likely to yield false discoveries.

Illustrative test family; BH adjustment

A	p = .010	q = .030	PASS
B	p = .040	q = .060	no
C	p = .200	q = .200	no

At q < 0.05, only A passes in this example. Repeated references to the same test count once; effect sizes stay unchanged.

ANALYSIS TOOLS

02 PANEL ANALYSIS + FEATURE OVERLAP

THE SAME 12 EXAMPLE RESTAURANTS

Region	Service time (seconds)	Matching CSAT (0-5)
Central	60, 68, 76, 84, 92, 100	3.5, 3.3, 3.4, 3.0, 3.1, 2.9
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10 within-between-correlation

Separates changes within the same restaurants over time from differences between restaurants.

Separate repeated-restaurant example; time effects off

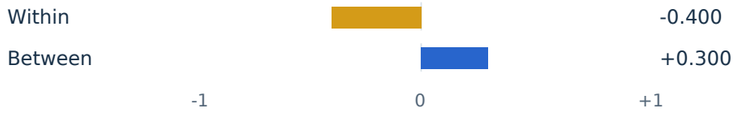


Within A and B, CSAT falls as service time rises. Between restaurants, B has longer times but higher CSAT.

11 within-between-regression

Estimates within- and between-restaurant slopes while considering several numeric drivers together.

Illustrative service-time slopes; control: staffing



Units: SD of adjusted CSAT per SD of adjusted driver

With staffing controlled, time has opposite slopes within vs between restaurants. Each model standardizes separately; these are not correlations.

12 correlation-matrix

Shows which numeric features move together, helping identify overlapping information.

12-restaurant calculation; Spearman r

	Time	Staff	CSAT
Time	1.000	-0.495	0.538
Staff	-0.495	1.000	0.438
CSAT	0.538	0.438	1.000

Time and staffing correlate at -0.495: slower restaurants tend to have fewer staff here. This checks overlap, not independent importance.